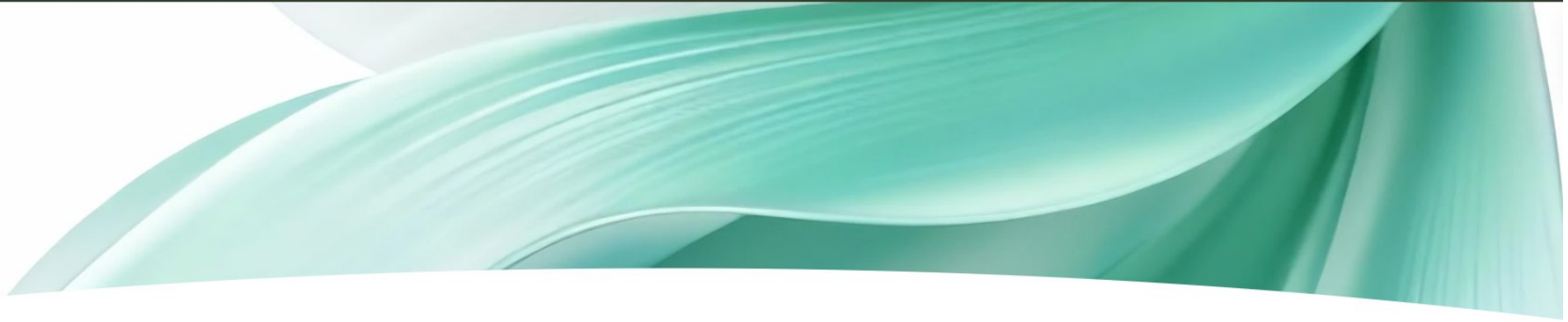
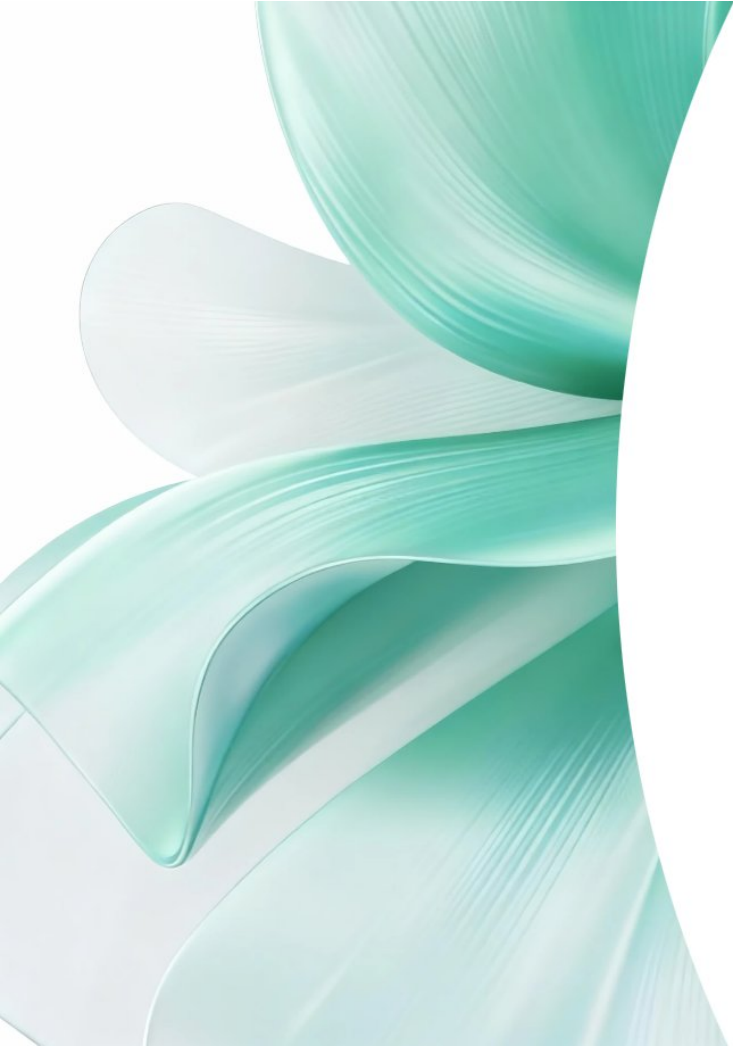




Supervised Machine Learning Models

Supervised machine learning models are a class of algorithms that learn from labeled training data to make predictions or classifications. By understanding the relationship between input features and output labels, these models can effectively analyze new data. Common types include linear regression, decision trees, and support vector machines. The success of supervised learning relies heavily on the quality and quantity of data provided for training.

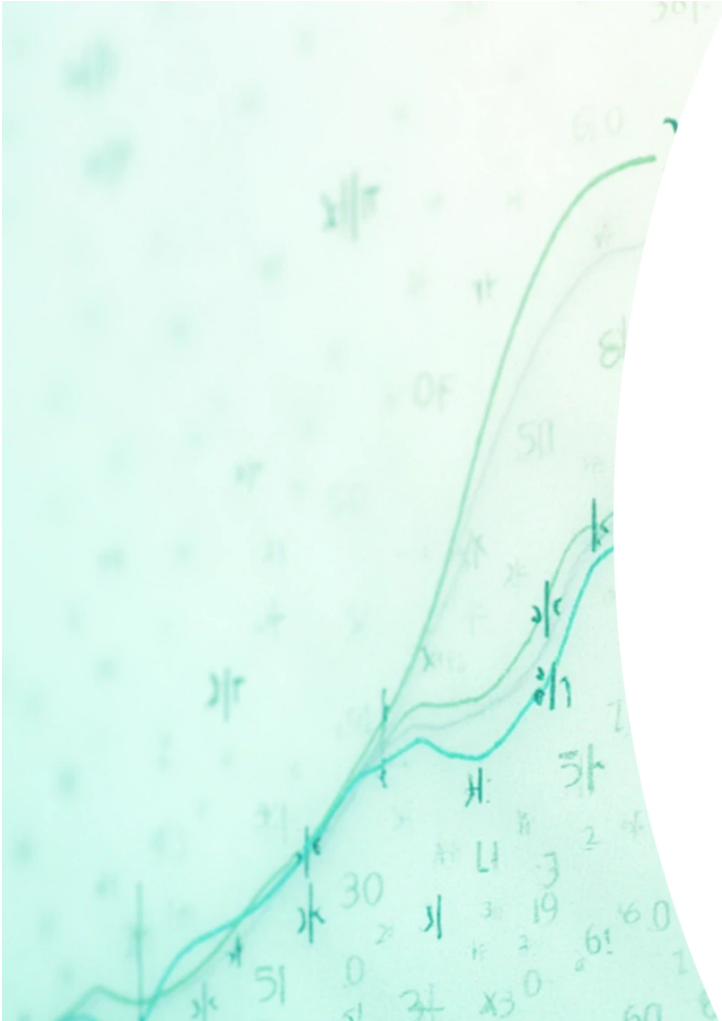


Logistic Regression

Logistic regression is a statistical method utilized for binary classification, predicting the probability of a given input belonging to a specific category.

It provides insights into the relationships between variables through its modeling capabilities.

It employs the logistic function to transform real-valued numbers into probabilities between 0 and 1, making it well-suited for binary dependent variables.



Mathematical Foundations

Logit Function

Transforms probabilities into odds using logarithm.

Model Equation

Expressed as: $\text{logit}(p) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$.

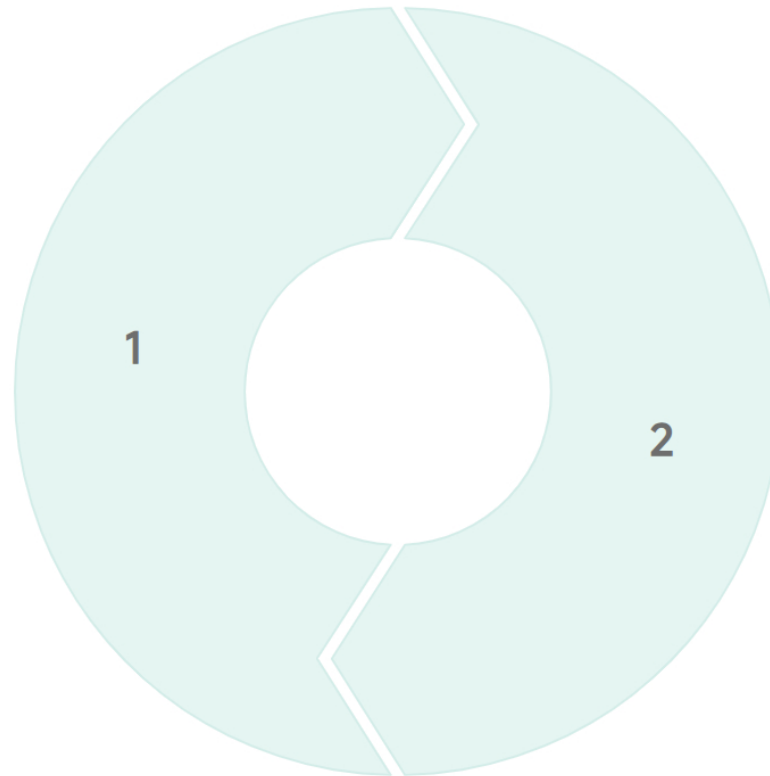
Estimation

Coefficients estimated using maximum likelihood estimation.

Limitations of Logistic Regression

Binary Focus

The model is mainly applicable to **binary outcomes**; while extensions exist, complex multi-class scenarios can be challenging.



Sensitive to Outliers

Extreme values in the dataset may disproportionately influence the results, leading to misleading interpretations.

Decision Tree Flow Model

1

Data Splitting

A decision tree splits data by asking **if-else** questions on features.

2

Purity Measures

It selects the feature and threshold that best separates data using Gini impurity or Entropy.

3

Leaf Nodes

When no more splits are possible, each leaf node gives a final prediction.

4

Interpretability

Highly interpretable; suitable for classification and regression but can **overfit**.



Random Forest Flow

1

Create many random trees

Multiple decision trees are trained on random bootstrap samples of the data, and at each split only a random subset of features is considered.

2

Each tree makes a prediction

Every tree independently predicts an output based on its own structure and sampled data.

3

Aggregate the results

The final prediction is made by majority voting (classification) or averaging (regression), which reduces overfitting and improves accuracy.

K-Nearest Neighbors (KNN) Classifier Flow

1

Store the training data

KNN does not learn a model; it simply stores all labeled data points in feature space.

2

Find the K nearest neighbors

It computes the distance to all training points and selects the K closest ones.

Support Vector Machine (SVM) — Flow

1

Map data into feature space

Data points are represented in a space for class separation; kernels map data into higher dimensions.

2

Find the optimal separating boundary

The hyperplane that separates classes with the maximum margin is identified.

3

Identify support vectors

The closest data points to the boundary influence the hyperplane's position.

4

Classify new data points

New points are classified based on their position relative to the hyperplane.

Gaussian Naive Bayes (Gaussian NB)

Gaussian Distribution

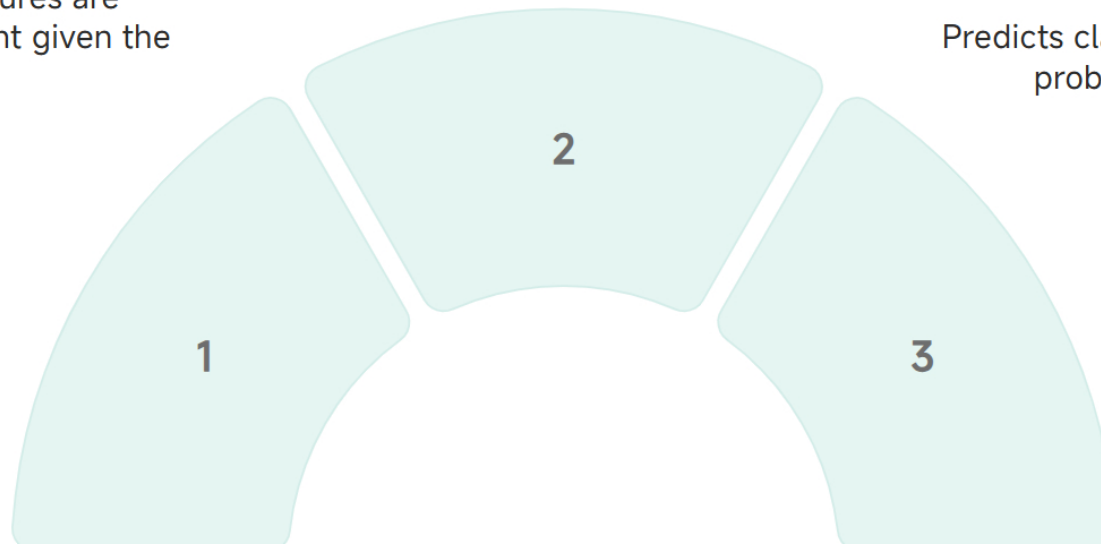
Models continuous features with Gaussian distribution for likelihood calculations.

Feature Independence

Assumes that all features are conditionally independent given the class.

Bayes' Theorem

Predicts class by calculating posterior probability for each class.



Unsupervised ML Models

1

Clustering

Groups similar data points together.

2

Dimensionality Reduction

Reduces the number of features while preserving essential information.

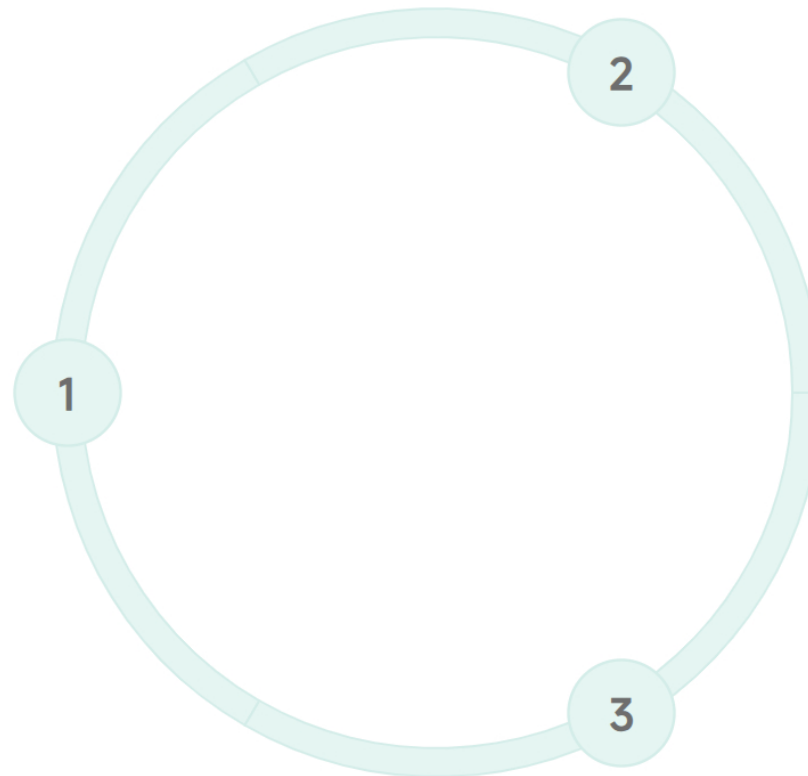
3

Anomaly Detection

Identifies unusual patterns that do not conform to expected behavior.

Understanding Linear Regression

Assume a linear relationship
Predict target variable y using a **linear combination** of input features.



Fit the model using training data

Learn weights that **minimize the mean squared error (MSE)** between predicted and actual values.

Predict new values

Compute output using the learned linear equation for new inputs.

K-Means Clustering Process

1

Initialize K centroids

Randomly pick K cluster centers in the feature space.

2

Assign points to nearest centroid

Each data point is assigned to the cluster with the closest centroid.

3

Update centroids

Recompute each centroid as the mean of all points assigned to that cluster.

4

Repeat until convergence

Iterate assignment → update until cluster assignments stop changing.

Hierarchical Clustering (Unsupervised, Agglomerative/Divisive)

1

Start with individual points

Each data point is treated as a single cluster (agglomerative) or all points in one cluster (divisive).

2

Compute pairwise distances

Use a distance metric (Euclidean, Manhattan) to find the closest clusters.

3

Merge or split clusters

Agglomerative: merge the closest clusters step by step.(**Bottom-UP**)

Divisive: split clusters iteratively.(**Top-Down**)

4

Build a dendrogram

The process continues until a single cluster (or K clusters) is reached; **Dendrogram** shows hierarchy.

DBSCAN (Density-Based Spatial Clustering)

1

Define parameters

Choose eps and min Pts.

2

Classify points

Identify core, border, and noise points.

3

Form clusters

Connect core points and neighbors.

4

Handle outliers

Label points not in any cluster as noise.

Mean Shift Clustering: Unsupervised Mode-Finding

1

Initialize Window

Start with a kernel/window around each data point.

2

Compute Mean

Shift the window towards the mean of points within it.

3

Iterate Until Convergence

Repeat the shift until the window stops moving.

4

Form Clusters

Points whose windows converge to the same mode are grouped into one cluster.

PCA (Principal Component Analysis, Dimensionality Reduction)

1

Standardize the data

Center and scale features.

2

Compute covariance matrix

Measure relationships between features.

3

Compute eigenvectors and eigenvalues

Identify principal components and variance explained.

4

Project data onto principal components

Reduce dimensions while retaining maximum variance.